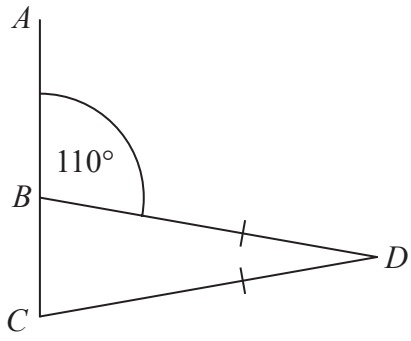
15



ABC is a straight line.
BD = *CD*
Angle *ABD* = 110°

Show that angle *BDC* = 40°

Give a reason for each stage of your working.

Angle CBD = 180° – 110° = 70°.

1 mark
Find CBD

Angles on a straight line sum to180°.

1 mark
Straight-line reason

BD = CD, so triangle BCD is isosceles.

Angle BCD = angle CBD = 70° (equal base angles).

Angle BDC = 180° – 70° – 70° = 40°.

1 mark
Full angle method

Angles in a triangle sum to180°.

1 mark
All reasons;40°

(Total for Question 15 is 4 marks)